

**MEDIATEK**

# MT8676 Yocto Sentry Mode User Manual

**Version:** 1.0  
**Release date:** 2024-09-19

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## Version History

| Version | Date       | Author  | Description      |
|---------|------------|---------|------------------|
| 1.0     | 2024-09-19 | Qing Wu | Official release |

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# Sentry Mode

## Overview

### Brief Introduction

The document introduces the implementation principles and operation methods of the Sentry Mode for MT8676. Sentry Mode refers to the real-time monitoring of the surroundings of the vehicle using external cameras when the vehicle is not in operation. It records video before and after detecting a threat (such as a collision or an anomaly detected by AI).

### Abbreviation

Table 1-1. Abbreviations

| Abbreviation | Expalnation                                |
|--------------|--------------------------------------------|
| APK          | Android Package Kit                        |
| APMCU        | Application Processor Microcontroller Unit |
| CAM_BE       | Camera Back End                            |
| CAM_FE       | Camera Front End                           |
| CAMSV        | Camera Subsystem                           |
| DRAM         | Dynamic Random-Access Memory               |
| GST          | GStreamer                                  |
| IPI          | Inter-Processor Interrupt                  |
| MDP          | Media Data Processor                       |
| MIPI         | Mobile Industry Processor Interface        |
| MIPICSI      | MIPI Camera Serial Interface               |
| PQDIP        | Picture Quality Digital Image Processor    |
| SCP          | System Control Processor                   |
| SoC          | System on Chip                             |
| V4L2         | Video for Linux 2                          |
| VDEC         | Video Decoder                              |
| VENC         | Video Encoder                              |

## 1.2 Architecture

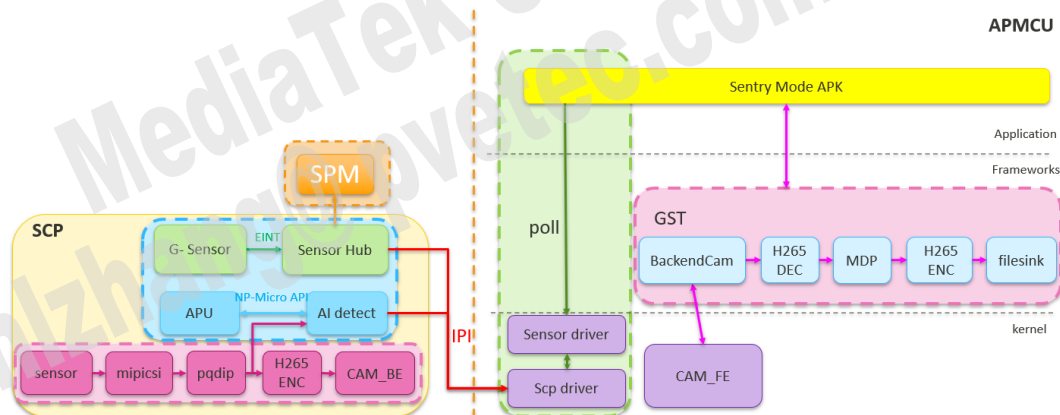


Figure 1-1. Sentry mode architecture

The architecture of Sentry Mode is divided into two parts: SCP and APMCU.

The SCP is a coprocessor on the System on Chip (SoC) that can operate at low power consumption while the system Central Processing Unit (CPU) is in sleep mode. The camera data collection, artificial intelligence (AI) detection, and collision detection functions for Sentry mode are all performed within SCP.

APMCU is the main processor of the SoC and remains in sleep mode during the detection process in Sentry Mode. When a threat is detected, the SCP wakes up the APMCU, and video recording is performed within the APMCU.

### 1.2.1 Hardware Path in SCP

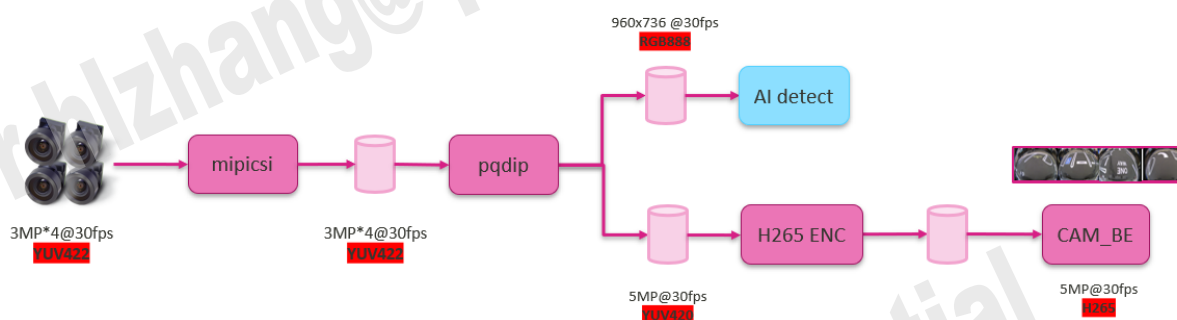


Figure 1-2. Sentry mode HW path in SCP

In SCP, the hardware path for Sentry mode is as shown in Figure 1-2.

The surround view camera is connected to MediaTek platform via serdes. MIPICSI receives mipi signal and then CAMSV (ISP P1 module) captures YUV422 data and stores it in DRAM. PQDIP (ISP P2 module) resizes the YUV422 data and converts its format to generate two streams. One stream is sent to AI detect module for AI detection, such as detecting if there are people nearby; the other stream is sent to video encoder module for data compression and storage. CAM\_BE is responsible for data storage updates and communication with APMCU.

## 1.2.2 Hardware Path in APMCU

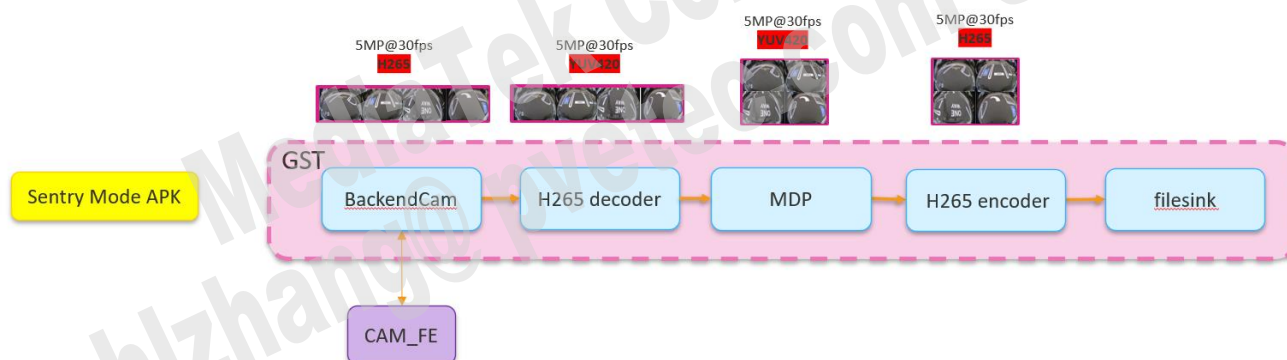


Figure 1-3. Sentry mode HW path in APMCU

In a hypervisor architecture, the APMCU for Sentry Mode operates on Yocto OS.

The main functions of the Sentry Mode APK include:

1. Enable or disable sentry mode
2. Detect g-sensor signal
3. Trigger video recording

Video recording is achieved through GST to create a pipeline, which includes CAM\_FE -> VDEC -> MDP -> VENC -> filesink. Communication between CAM\_FE and the CAM\_BE module in SCP via IPI is established to acquire camera images. The image layout is in a landscape format with H265 encoding. First, the video decoder (VDEC) converts the format to YUV420. Then, the MDP changes the image layout to a grid format. Finally, the video encoder (VENC) re-encodes the images into H265 format and stores them as an MP4 file.

## 1.3 SW Control Flow

### 1.3.1 Enable Sentry Mode

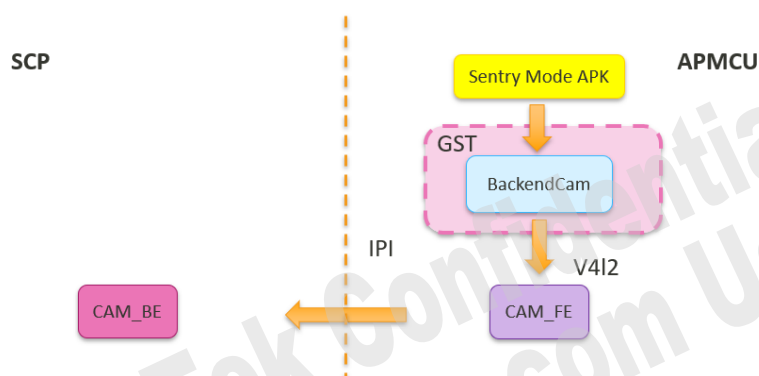


Figure 1-4. Sentry mode enable flow

Through the V4L2 interface, the application sends the sentry mode on/off settings to CAM\_FE. Upon receiving the command, CAM\_FE passes the settings to CAM\_BE on the SCP side via IPI. After receiving the command, CAM\_BE performs the corresponding operations to enable or disable sentry mode.

### 1.3.2 Low Power Mode

Upon receiving the sentry mode on message, the pipeline in SCP is established as depicted in Figure 1-2 for saving the camera images and detecting the threats.

### 1.3.3 Wake Up Mode

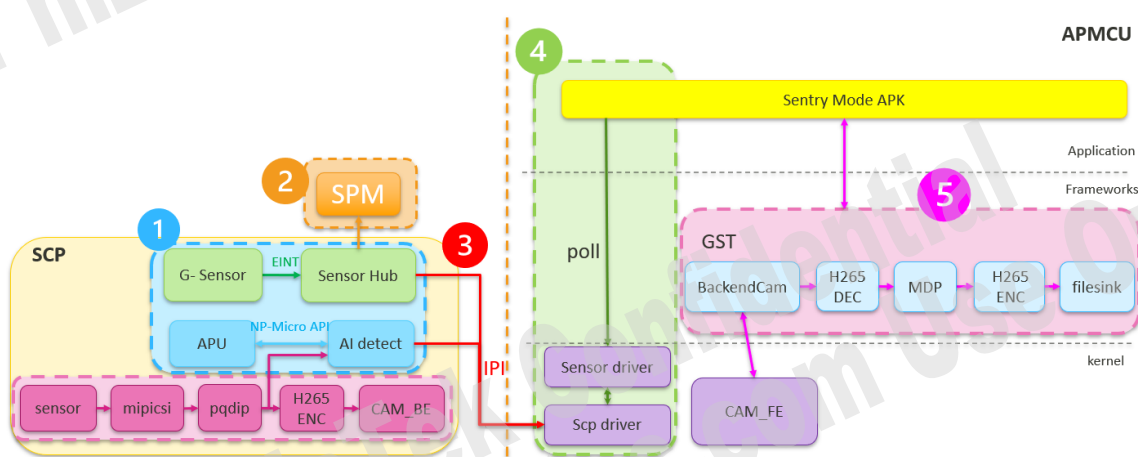


Figure 1-5. Sentry mode wake up flow

#### 1. Threat Detection:

- G-sensor detects collision or AI detects a person or an object approaching the vehicle.

#### 2. SCP Analysis and Wake-Up:

- SCP Analysis: The System Control Processor (SCP) analyzes the detected threat or anomaly.
- Send Wake-Up Event: The SCP sends a wake-up event to the System Power Management (SPM).
- SPM Power-On: The SPM powers on the Application Processor Microcontroller Unit (APMCU, ARM) and performs a kernel resume.

#### 3. Message Transmission:

- SCP Transmits Message: The SCP transmits the collision message to the APMCU via Inter-Processor Interrupt (IPI) using the SCP Driver.

#### 4. Event Polling:

- Application Polling: The Sentry Mode application (Sentry Mode APK) polls for G-sensor events to confirm the threat or anomaly.

#### 5. Video Recording:

- Create Data Processing Pipeline: The Sentry Mode application creates a data processing pipeline using GStreamer (GST).
- Start Video Recording: The application starts video recording to capture footage before and after the detected threat or anomaly.

### 1.3.4 IPI Communication between APMCU and SCP

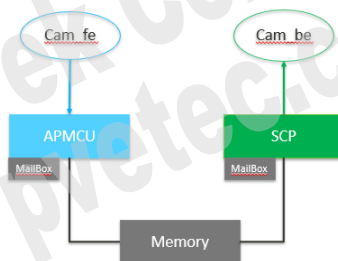


Figure 1-6. IPI Communication between APMCU and SCP

In Sentry Mode, the APMCU and SCP communicate via the mailbox. This includes settings for enabling or disabling Sentry Mode, as well as enqueueing and dequeueing camera image buffers

### 1.3.5 Memory Sharing between APMCU and SCP

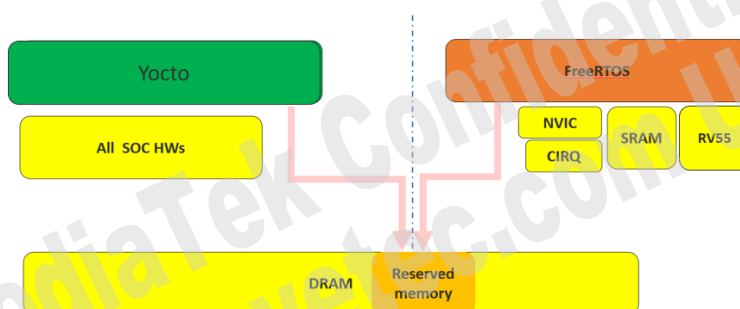


Figure 1-7. Memory sharing between APMCU and SCP

As shown in Figure 1-7, a portion of Dynamic Random-Access Memory (DRAM) is reserved for use by Sentry Mode. In this way, in low-power mode, the SCP stores camera images in the reserved memory which can also be accessed by Yocto. When an anomaly is detected, Yocto can directly access this buffer and record it as an MP4 file, without the need for buffer copying between FreeRTOS and Yocto.

## 1.4 How to Run Sentry Mode

### 1.4.1 Sensor

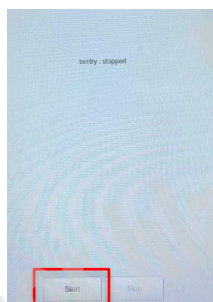
In Sentry Mode, an AVM YUV sensor needs to be connected to the SPM8676 reference board. Specifically, a MAX96712 camera daughterboard should be connected to the CON12 slot on the SPM8676 reference board, and four YUV sensors should be connected to the CON501 slot on the daughterboard.

### 1.4.2 App Setting

Enter the following command in the Yocto shell interface:

```
C:\Users\mtk40434>adb shell
sh-3.2# source /data/wayland_env_file
sh-3.2# sentry
```

The sentry mode application interface shown as below will pop up, allowing you to click start/stop to enable or disable the sentry mode.



### 1.4.3 Recording

In the event of a collision, 10 seconds of data before and after the collision will be recorded and stored in the /data/ directory.

- Path: /data
- Name: *sentry\_data\_year\_month\_day\_hour\_minute\_second.mp4* (Correct time can only be obtained when the platform is connected to the network, otherwise the default time of the platform will be used.)
- Format: MP4
- Duration: 20s
- Video Layout: Horizontal strip



If you need to display it in a grid format, you need to enter the following command.

```
adb shell setprop vendor.sentrydvr.changepipeline.enable 1
```



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